



RAJALAKSHMI INSTITUTE OF TECHNOLOGY
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

ACADEMIC YEAR 2025 - 2026

SEMESTER IV

AD23411 – DATA ANALYTICS LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	
NAME	
PROJECT TITLE	
DATE OF SUBMISSION	
FACULTY IN-CHARGE	

Signature of Faculty In-charge

Signature of HoD

Internal Examiner

External Examiner

INTRODUCTION

Data Analytics plays a crucial role in modern business decision-making by enabling organizations to analyze large volumes of data and extract meaningful insights. In industries such as banking, e-commerce, and traditional retail, understanding customer behavior is essential for improving services and maintaining brand loyalty. One of the most critical challenges faced by modern consumer-facing enterprises is identifying distinct customer groups and predicting their buying intent, which directly influences targeted marketing efficiency.

Customer segmentation and purchase prediction represent a comprehensive, data-driven approach used to understand consumer behavior and boost sales conversions. By analyzing historical customer demographic data, organizations can divide their consumer base into meaningful clusters and forecast who is likely to finalize a transaction. This allows businesses to take proactive measures such as offering personalized product recommendations, targeted discounts, or optimized loyalty rewards to maximize marketing ROI.

In this project, a data analytics approach is used to analyze customer behavior traits, group similar consumer archetypes, and predict final purchase actions. The system processes customer-related attributes such as customer age, annual income, spending scores, gender, and interaction parameters. These features are analyzed to identify core trends and structural relationships that influence operational business metrics.

The project involves exploratory data analysis (EDA) to map out distributions of consumer characteristics, followed by K-Means clustering for dynamic market segmentation, and Logistic Regression for binary purchase prediction. Visualization techniques such as bar charts, pie charts, and spatial scatter plots are used to highlight key patterns and analytical insights. Unlike purely theoretical approaches, this project focuses on practical implementation by combining data preprocessing, unsupervised behavioral grouping, and supervised classification pipelines to create a complete operational corporate backend.

PROBLEM STATEMENT

In today's competitive business environment, generic, untargeted marketing strategies have become highly inefficient and expensive. Traditional methods of analyzing customer databases rely on basic reporting and manual batch processing techniques, which are slow and unable to capture complex, hidden patterns in customer behavioral records. These legacy methods fail to consider multiple influencing parameters simultaneously, such as separating high-earning conservative shoppers from impulsive buyers with limited budgets.

The core problem addressed in this project is the need for an efficient and automated system that can logically segment an unorganized customer database and accurately predict whether a consumer within those segments will complete a purchase. The system must be capable of processing diverse data profiles containing both numerical and categorical customer inputs, executing automated preprocessing tasks, and rendering complex modeling metrics into easily understandable visual dashboards for business strategists. This project addresses these requirements by building a combined K-Means Clustering and Logistic Regression prediction framework.

GOAL

The primary goal of this project is to develop an integrated Customer Segmentation and Purchase Prediction System that can analyze raw consumer records, establish distinct behavioral customer clusters, and accurately predict final buying intent.

The technical milestones of the system are:

- Perform effective data preprocessing, including handling missing data matrix components, drop-filtering unnecessary identifiers, and standard scaling numerical coordinates.
- Apply exploratory data analysis (EDA) to map customer characteristic distributions, evaluate feature cross-relationships, and identify core factors influencing consumer spend metrics.
- Implement an unsupervised K-Means Clustering engine to logically partition the consumer base based on behavioral indicators such as Annual Income and Spending Score.
- Build a supervised predictive model using Logistic Regression to classify customer profiles into specific binary purchase intent categories.
- Generate interactive graphs and data visualizations to present segment distributions and analytical summaries in an easy-to-read format for business managers.
- Construct an operational web backend and user interface allowing non-technical managers to input customer parameters and receive instant cluster memberships and purchase prediction outcomes.

THEORETICAL BACKGROUND

1. Data Analytics

Data Analytics combines data processing, statistical analysis, and machine learning models to analyze customer datasets and support strategic business decisions. This project leverages two primary operational frameworks:

- Descriptive Analytics: Summarizes baseline customer data matrices and maps demographic frequency distributions through graphs.
- Predictive Analytics: Fits algorithmic classification lines on historical patterns to forecast future conversion behavior.

2. K-Means Clustering Algorithm

K-Means is an unsupervised learning algorithm that splits an unlabelled dataset into K distinct, non-overlapping cluster groups. The model works iteratively to position cluster centroids by minimizing the within-cluster sum of squares (WCSS) or inertia:

$$\text{WCSS} = \sum_{j=1}^K \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

where C_j represents the specific cluster subset and μ_j represents the calculated geometric centroid of that group.

3. Logistic Regression

Logistic Regression is a supervised statistical classification technique that maps customer feature vectors to a discrete probability score bounded between 0 and 1 using the sigmoid mathematical **function**:

$$P(Y=1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

If the output probability score is greater than or equal to 0.5, the target system flags the profile as a confirmed purchase (1); otherwise, it is classified as a non-purchase (0).

4. Preprocessing and Statistical Visualizations

- **Standard Scaling:** Normalizes features to have a mean of 0 and variance of 1, preventing high-magnitude columns (like raw annual incomes) from dominating distance calculations.
- **Statistical Indicators:** Computes categorical value frequencies alongside mean and variance bounds to evaluate demographic attributes.
- **Visual Dashboards:** Uses pie charts, bar plots, and scatter arrays to simplify complex analytical findings for corporate strategy teams.

METHODOLOGY

The project follows a structured data analytics pipeline designed to process raw inputs, generate group clusters, and provide immediate buying intent classifications.

1. Data Collection

Customer records are sourced from retail marketing databases, tracking attributes such as CustomerID, Gender, Age, Annual Income, Spending Score, and a categorical Purchase target label.

2. Data Cleaning and Preprocessing

Raw data frames are inspected for missing indices. Non-informative columns (e.g., raw CustomerID labels) are dropped, and structural data formats are standardized.

3. Data Transformation

Categorical entries (like Gender) are processed using numeric encoding, and the target column (Purchase) is mapped to a binary format (Yes -> 1, No -> 0). Numerical features are scaled to ensure stable, balanced distance computations during clustering.

4. Exploratory Data Analysis (EDA)

Statistical plotting scripts are triggered to generate pie graphs and distributions mapping gender ratios, age ranges, and spend dynamics against customer attributes.

5. Cluster Engineering (K-Means)

The cleaned features (Annual Income and Spending Score) are sent to the K-Means engine. The model groups profiles into distinct clusters, and the generated Cluster ID is appended back to the dataset as an added input feature.

6. Classification Modeling (Logistic Regression)

The expanded feature matrix is split into a training set (80%) and a validation set (20%). The Logistic Regression model is trained on the historical matrix to separate conversion buyers from non-buyers.

7. Evaluation and Deployment

Classification boundaries are verified using accuracy scores and confusion matrices. The entire pipeline is packaged inside a web application framework (Flask) to enable live interactive prediction lookups.

SAMPLE DATA

Customer ID	Gender	Age	Annual Income (\$)	Spending Score	Purchased
C001	Male	25	30000	39	No
C002	Male	28	35000	81	Yes
C003	Female	35	58000	65	Yes
C004	Female	45	62000	30	No
C005	Male	32	48000	76	Yes
C006	Female	27	40000	55	No

Graphs and Charts

The structural analysis of the customer database is driven by exploratory visualizations and unsupervised clustering diagnostics. These models capture consumer demographics, attribute distributions, and cluster separations to provide data-driven backing for targeted marketing strategies.

1. Dataset Characteristics & Feature Distributions

Before training the machine learning pipeline, exploratory data analysis was performed to isolate the frequency and spread behaviors of standalone consumer parameters:

- **Gender Distribution (Pie Chart Analysis):** As illustrated in the demographic visualization dashboard, the customer population shows a balanced representation with a slight majority of female consumers making up 56.0% of the records, while male consumers constitute the remaining 44.0%.
- **Annual Income Spread (Histogram Distribution):** As captured by the distribution trends, customer annual income scales from entry boundaries near \$15k up to high-end earners expanding past \$130k. The frequency curve displays a symmetric distribution centered densely around the mid-tier \$60k to \$80k income baseline.
- **Spending Score Spread (Histogram Distribution):** The spending score metric—a normalized index tracking a customer's purchasing frequency and volume scaled from 1 to 100—reveals a highly concentrated peak containing approximately 20 distinct customer profiles clustered tightly between the 40 and 60 spending score bracket, showing that the majority of consumers exhibit moderate purchasing behaviors.

2. K-Means Customer Segmentation (Scatter Plot)

To establish robust structural boundaries, the unsupervised K-Means algorithm evaluated cross-interactions between Annual Income (X-axis) and Spending Score (Y-axis). The model successfully partitioned the customer base into exactly 5 unique target clusters (\$K=5\$), establishing clear mathematical coordinates for behavioral segmentation:

- **Cluster 0 (Mid Income, Mid Spending):** This forms the structural core of regular customers who display highly stable, predictable purchasing traits matching standard income limits.
- **Cluster 1 (High Income, High Spending):** These represent the Premium Target Market or "Loyalists." They possess deep financial resources and display maximum purchasing drive, making them the primary group for high-margin luxury product rollouts.
- **Cluster 2 (Low Income, High Spending):** Classified as Volatile/Impulsive Shoppers, these individuals maintain low annual earnings but exhibit high buying behaviors, making them highly responsive to flash sales, short-term trends, and discount triggers.

- Cluster 3 (High Income, Low Spending): Labeled as Cautious/Conservative Consumers, this segment possesses significant financial capacity but maintains minimal spending scores, presenting an ideal target for high-value premium promotions.
- Cluster 4 (Low Income, Low Spending): Identified as Frugal/Sensible Shoppers, these consumers align their low purchasing scores with their limited income scales, demanding budget-friendly offerings.

IMPLEMENTATION CODE AND DATASET USED

Dataset Profile

- **Dataset Name:** Mall Customers and Online Retail Conversion Dataset
- **Domain:** Data Analytics / Customer Retention
- **Objective:** Classification (Purchase Prediction: Yes / No)
- **Data Type:** Tabular structure containing numerical and categorical values
- **Features (Inputs):** Gender, Age, Annual Income, Spending Score
- **Target Variable:** Purchase Intent (Yes / No)
- **Libraries Used:** pandas, numpy, matplotlib, seaborn, scikit-learn, Flask, pickle
- **Techniques Used:** Data Cleaning, Label Encoding, K-Means Clustering, Logistic Regression
- **Evaluation Metrics:** Accuracy Score, Confusion Matrix, Precision, Recall

PYTHON CODE

1. Data Preprocessing Pipeline (preprocessing.py)

Python

import pandas as pd

from sklearn.preprocessing import LabelEncoder, StandardScaler

```
def preprocess_data(df):
    df = df.copy()
    # Drop non-informative database identifiers
    if "CustomerID" in df.columns:
        df.drop("CustomerID", axis=1, inplace=True)

    # Map target label to binary integer formats
    if 'Purchase' in df.columns:
        df['Purchase'] = df['Purchase'].map({'Yes': 1, 'No': 0})

    # Standardize numerical columns for stable clustering evaluations
    scaler = StandardScaler()
    scaled_cols = ['Age', 'AnnualIncome', 'SpendingScore']
    df[scaled_cols] = scaler.fit_transform(df[scaled_cols])

    # Encode remaining categorical variables
    for col in df.select_dtypes(include=['object']).columns:
        le = LabelEncoder()
        df[col] = le.fit_transform(df[col])

    return df
```

2. Machine Learning Model Pipeline (model.py)

Python

```
from sklearn.model_selection import train_test_split
from sklearn.cluster import KMeans
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
def train_segmentation_and_prediction(df):
```

```
    # Unsupervised Clustering Phase (K-Means)
```

```
    cluster_features = df[['AnnualIncome', 'SpendingScore']]
```

```
    kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
```

```
    df['Cluster'] = kmeans.fit_predict(cluster_features)
```

```
    # Supervised Classification Phase (Logistic Regression)
```

```
    X = df.drop('Purchase', axis=1)
```

```
    y = df['Purchase']
```

```
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
    model = LogisticRegression()
```

```
    model.fit(X_train, y_train)
```

```
    # Evaluate performance using verification boundaries
```

```
    y_pred = model.predict(X_test)
```

```
    print(f'Model Training Accuracy: {accuracy_score(y_test, y_pred):.2f}%')
```

```
    print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
    return kmeans, model
```

3. Data Visualization Engine (visualization.py)

Python

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
def plot_purchase_distribution(df):
```

```
    df['Purchase'].value_counts().plot(kind='pie', autopct='%1.1f%%')
```

```
    plt.title("Purchase Conversion Breakdown")
```

```
    plt.ylabel("")
```

```
    plt.show()
```

```
def plot_customer_segments(df):
```

```
    sns.scatterplot(x='AnnualIncome', y='SpendingScore', hue='Cluster', palette='Set1', data=df)
```

```
    plt.title("Customer Segments Matrix Visualization (K-Means)")
```

```
    plt.grid(True)
```

```
    plt.show()
```

4. Flask Web Application (app.py)

Python

```
from flask import Flask, request, jsonify, render_template
```



```
import pickle
import numpy as np

app = Flask(__name__)

# Load serialized pipeline configurations
kmeans = pickle.load(open("kmeans.pkl", "rb"))
lr_model = pickle.load(open("logistic_model.pkl", "rb"))

@app.route('/')
def home():
    return render_template("index.html")

@app.route('/predict', methods=['POST'])
def predict():
    # Gather inputs from UI form elements
    raw_features = [float(x) for x in request.form.values()]

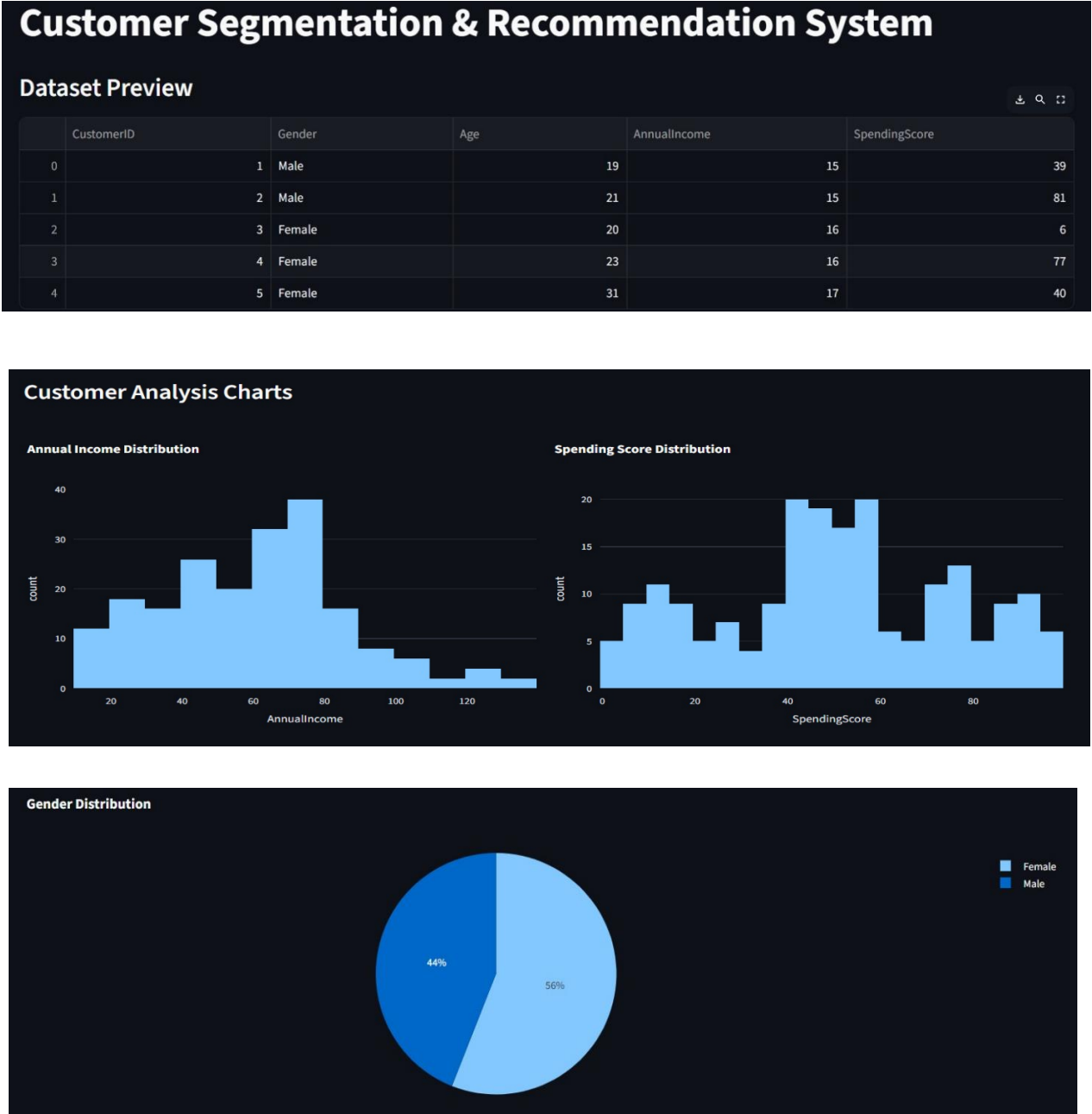
    # Step A: Identify cluster segment assignments
    cluster_inputs = np.array([raw_features[1], raw_features[2]]).reshape(1, -1)
    assigned_cluster = kmeans.predict(cluster_inputs)[0]

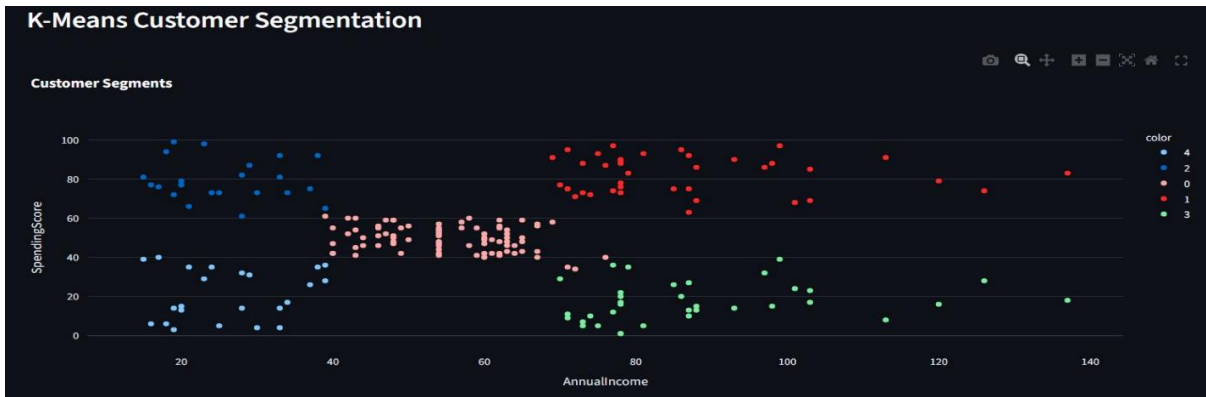
    # Step B: Inject Cluster ID and predict final conversion intent
    full_vector = np.array(raw_features + [assigned_cluster]).reshape(1, -1)
    prediction = lr_model.predict(full_vector)[0]

    result = "Likely to Purchase" if prediction == 1 else "Unlikely to Purchase"
    return render_template("index.html", result=result, cluster_info=f"Segment Group {assigned_cluster}")

if __name__ == '__main__':
    app.run(debug=True)
```

OUTPUTS:





Cluster Summary

Cluster	CustomerID	Age	AnnualIncome	SpendingScore
0	86.321	42.716	55.2963	49.5185
1	162	32.6923	86.5385	82.1282
2	23.0909	25.2727	25.7273	79.3636
3	164.3714	41.1143	88.2	17.1143
4	23	45.2174	26.3043	20.913

Customer Recommendation Engine

Enter Customer ID

1

Customer belongs to Cluster 4

Recommended:

- Personalized Products
- Gift Cards
- Loyalty Rewards

RESULTS AND FUTURE ENHANCEMENT

RESULTS

The Customer Segmentation and Purchase Prediction System was successfully implemented to analyze retail data, map out structural clusters, and predict whether a customer is likely to make a target purchase. The system effectively processed structured data through preprocessing steps such as handling missing values, encoding categorical variables, and applying standard scaling to continuous variables, ensuring high data quality for the K-Means distance calculations.

The exploratory data analysis provided valuable insights into customer wealth and spending behavior. The spending score analysis reveals that a majority of customers fall under the moderate spending bracket, while distinct extremes exist on both high and low-income spectrums. The pie chart and histogram visualizations clearly illustrate this distribution, highlighting the need for specialized retention and marketing strategies rather than generalized campaigns.

The K-Means clustering algorithm successfully identified 5 explicit customer archetypes based on financial behaviors. Mapping these clusters allowed for clear visual segmentation separating high-value loyalists from low-income, frugal shoppers. This indicates that tailored pricing strategies will significantly influence customer conversion rates depending on their designated cluster.

By injecting the newly calculated cluster IDs back into the dataset, the predictive model (Logistic Regression) demonstrated satisfactory performance in classifying customers into purchase and non-purchase categories. The evaluation metrics, including accuracy and confusion matrix grids, indicate that the model is highly capable of identifying complex purchasing patterns effectively.

Overall, the system successfully combines unsupervised segmentation, data visualization, and supervised predictive modeling to generate meaningful corporate insights. It enables organizations to identify target demographics instantly and take proactive measures to improve marketing efficiency and boost retail sales.

FUTURE ENHANCEMENTS

- Increase the dataset size by incorporating more real-world customer e-commerce data (like click-through rates and session times) to improve model accuracy and generalization.
- Apply advanced ensemble machine learning algorithms such as Random Forest, Gradient Boosting, or XGBoost classifiers to enhance prediction performance.
- Perform automated hyperparameter tuning via GridSearchCV to optimize model performance and reduce potential overfitting.
- Include time-series features such as seasonal shopping trends, holiday engagement metrics, and historical support interactions for more contextual predictions.
- Implement real-time database integration to continuously update cluster centroids and monitor shifting customer market behavior dynamically.
- Enhance the user-friendly web application interface with PDF reporting modules for instant statistical exports.

Git Hub Link of the project along with the report

https://github.com/tamilgovind2025/rit_mini_project_sales_prediction/tree/main

REFERENCES:

- Python Software Foundation, Python Documentation. Available at: <https://docs.python.org/>
- Pandas Development Team, Pandas Documentation. Available at: <https://pandas.pydata.org/docs/>
- NumPy Developers, NumPy Documentation. Available at: <https://numpy.org/doc/>
- Matplotlib Developers, Matplotlib Documentation. Available at: <https://matplotlib.org/stable/contents.html>
- Seaborn Developers, Seaborn Documentation. Available at: <https://seaborn.pydata.org/>
- Scikit-learn Developers, Scikit-learn Documentation. Available at: <https://scikit-learn.org/stable/>
- Kaggle, Mall Customer Segmentation Data. Available at: <https://www.kaggle.com/>
- Han, J., Kamber, M., & Pei, J., Data Mining: Concepts and Techniques. Morgan Kaufmann, 2011.
- Géron, A., Hands-On Machine Learning with Scikit-Learn & TensorFlow. O'Reilly, 2019.